Paper review · Visionary

Teacher recoverability in DAgger for VLAs

DAgger: Ross, Gordon and Bagnell, AISTATS 2011 (arXiv:1011.0686).

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When is a VLA a good DAgger expert?

DAgger’s bound has an expert term

J(DAgger policy) ≤ J(expert) + u·T·ε + O(1) (Theorem 3.2).

u bounds how much the expert’s cost increases after one mistake of the learner. u is small only when the expert recovers in a few steps.

The gap

VLA-OPD and Multi-Teacher OPD let a small VLA act and label its states with VLA teachers. Reward-DAgger kept only the 4 LIBERO-10 tasks on which its π0.5 expert recovers.

None of them reports how often the teacher recovers from the student’s states.

Hypotheses

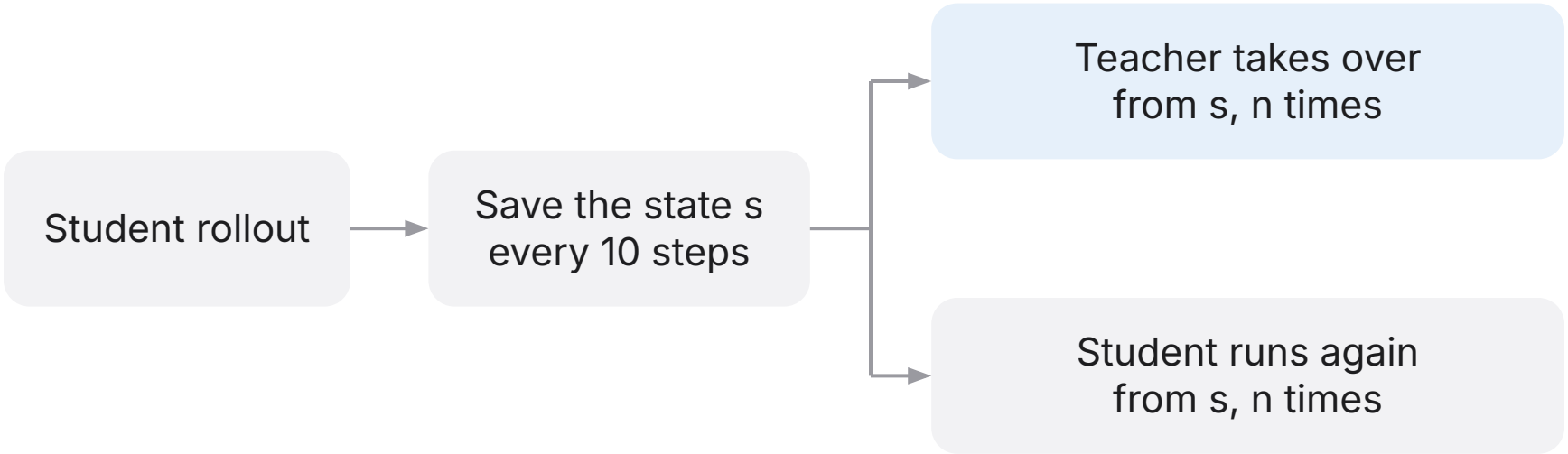
H1 From the student's states, the teacher succeeds less often than from the task start. This rate predicts the DAgger gain per task.

H2 At a fixed label budget, labels only on states that the teacher recovers from give higher student success than labels on all states.

H3 On-policy labels beat offline distillation only when the student is too small to copy the teacher.

If H1 holds, the labels of a model teacher need a check before use. If H2 holds, the same check also saves labels.

Measuring u: the takeover test



The recovery rate r(s) is the share of teacher takeovers that succeed. The student's re-runs show how often s succeeds without help.

Design

Teacher	π0.5 LIBERO checkpoint (openpi; 96.85% on LIBERO), frozen.
Students	SmolVLA (450M); π0 with LoRA (larger); a diffusion policy (smaller).
Round 0	Behavior cloning on 10% of the LIBERO demonstrations.
Conditions	Offline distillation; DAgger; DAgger on recoverable states; takeovers.
Tasks	LIBERO, 4 suites; LIBERO-Plus camera and layout shifts.
Metrics	Success per teacher label; r(s) per task; offline vs on-policy gap.
Controls	Same label budget; 3 rounds; paired seeds; student re-runs.

Compute and risks

Compute (estimate)

Each round: 2,000 student episodes, and takeover tests with re-runs on 2,000 saved states.

About 60 GPU-hours of rollouts, 50 of fine-tuning and 30 of evaluation: 140–180 GPU-hours on one RTX 5090.

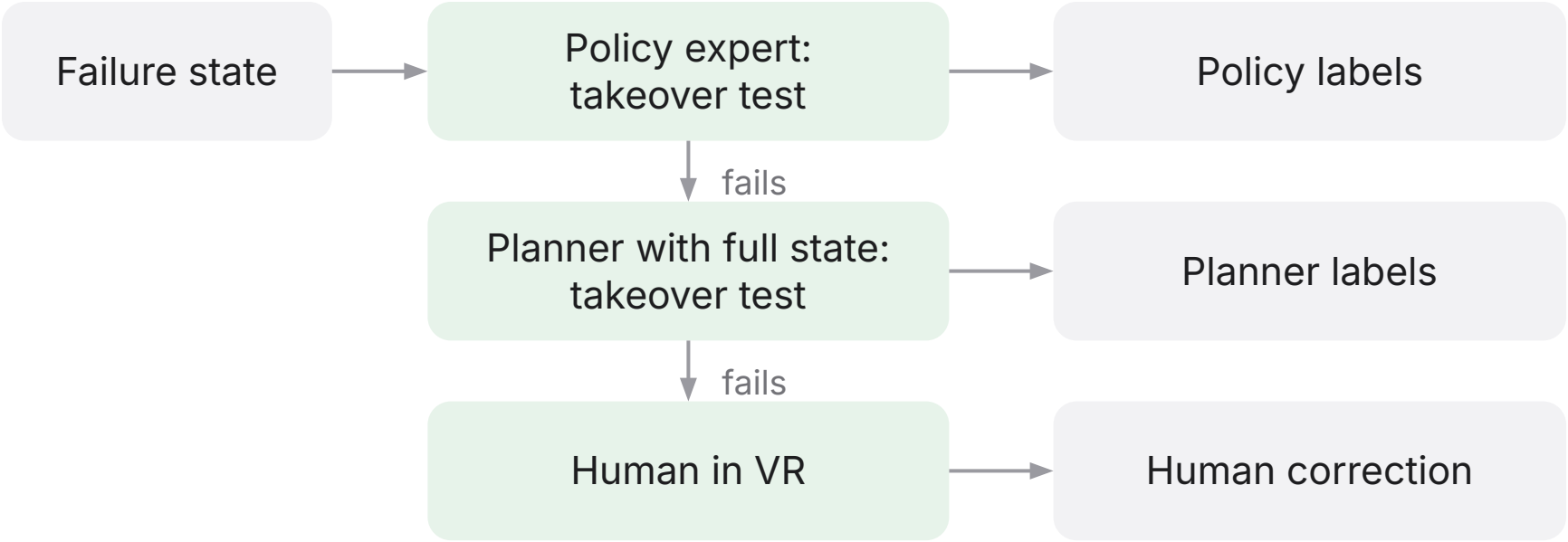
Risks

The teacher is strong on LIBERO, so few states may be unrecoverable. LIBERO-Long and LIBERO-Plus give more.

The test needs an exact state reset, which real robots do not have.

FastOPD already distills π0.5 into SmolVLA from the dataset. The contribution is the measurement and the routing.

Application: a failure router



A failure goes down the list until an expert recovers from it.

What a user gets, and the gaps

For long-horizon household tasks

- Recovery data for each failure type.
- Human time only on the failures that no model recovers from.
- A list per task of which expert fixes which failures.

Prior work

HG-DAgger and RaC use a person for each correction. VLA-OPD uses a model teacher for every state.

The 2025 BEHAVIOR winner had no recovery demonstrations. It used hand-written correction rules.

Pilot first, then the router

Pilot (go/no-go)

LIBERO-Long, the round-0 SmolVLA student, 300 saved states from failed episodes.

The work continues only if the teacher recovers from clearly fewer of them than its success from the start (H1).

Router test

2–3 household tasks in simulation with a policy expert, a planner with full state and a person in VR.

Measures: student success per expert-minute, and the share of failures that each expert recovers from.

What this builds on

Work	What it does	What this proposal adds
DAgger, 2011	Expert labels all states; bound has u.	u for a VLA teacher.
HG-DAgger, RaC	A person takes over, then corrects.	People only where models fail.
VLA-OPD, 2026	A VLA teacher labels student states.	A test of those labels.
Reward-DAgger	A reward model gates π0.5 takeovers.	Which states π0.5 can fix.
Corrector vs retreat	30% of failures succeed on a re-run.	Re-runs as a control.

Sources

1. Ross, Gordon and Bagnell. DAgger, AISTATS 2011: Theorem 3.2, Section 2.2.
2. Kelly et al. HG-DAgger, 2019. Hu et al. RaC, 2025.
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6. Foster, Block and Misra. Is Behavior Cloning All You Need?, 2024.
7. Physical Intelligence. π0.5, 2025, and the openpi LIBERO checkpoint. Shukor et al. SmolVLA, 2025.
8. Liu et al. LIBERO, 2023. LIBERO-Plus, 2025.
9. Larchenko et al. 1st place, 2025 BEHAVIOR Challenge, 2025.